

Xiaoxiao YU

Agent Engineer | Full-Stack Engineer

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Full-stack agent engineer who ships production AI products end to end, from React/TypeScript frontends through Python and Node.js backends to the agent runtime underneath, often as the sole engineer. At Viciking since July 2026, built four agent products, used by dozens of internal users, that **doubled video-production efficiency**, with **~90% of automated outputs publish-ready**. Before: a browser-agent startup, zkWasm apps with thousands of users, and React Native apps.

Experience

Viciking (Beijing Weixi Network Technology)

Jul 2026 – Present

Agent Engineer | sole engineer, frontend to agent runtime, on four internal AI production products

- Business impact vs. manual editing: **2× production efficiency**, with **~90% of automated outputs publish-ready**; dozens of internal users across multiple business lines.
- WeiChuang Drama Studio** (React 19, FastAPI, Temporal, PostgreSQL): script → storyboard → video workbench (agent and manual modes, team permissions, CapCut-draft export). In agent mode, a 1-minute episode goes **from script to finished video in 15–30 min**, with episodes running in parallel, and **~90% are publish-ready** thanks to model routing and fallback, voice repair, storyboard rules against empty shots, and auto subtitles. Built a resumable coordinator (frozen inputs, idempotent compare-and-set writes, classified retries, partial-success reruns) and **per-call model traces** (redacted I/O, cached tokens, stream checkpoints, failure evidence) to diagnose slow or stuck calls. 1,890 commits; ~5,000 tests plus Playwright end-to-end.
- Weixi Studio** (TypeScript, React, Remotion, PostgreSQL, MCP): turned OpenChatCut (local-first, single-user open-source agentic video editor, 2.2k ★) into a multi-tenant SaaS: workspaces, roles and feature grants, a shared asset library, resumable bulk browser-to-cloud upload, a template-driven production center, and separate agent/generation/export workers; ~1,500 commits. Built a durable agent runtime: Postgres leases with fencing tokens, checkpointed turns, human-decision states, and per-call idempotency keys; paid or irreversible tools escalate to a human instead of replaying. Added a session-bound, revision-safe MCP server so external agents edit the same timeline.
- Creative Engine** (TanStack Start, pgvector): writer agent with 9 least-privilege tools over frozen, provenance-labeled context; 14 ablation/LLM-judge experiments decided which prompt mechanisms stayed.
- Screenplay Agent Studio** (FastAPI, Bun, Pi agent SDK, OpenTelemetry): novel-to-series writing with 6 stage contracts and 13 role agents behind per-stage tool allowlists; cache-stable versioned prefixes; workspace files as the single source of truth, re-attached after compaction; human-approved atomic edits. **OpenTelemetry metrics and alerts** on agent-run failure ratio and stalled human decisions, plus an append-only audit log. My eval harness showed specialist delegation cost **3.7× tokens for no quality gain**, so production stayed single-lead.

Resona

May 2026 – Jun 2026

Founder & sole engineer | URL in, product demo video out | Ant Group MaShang AI Accelerator; trial customers

- Built the browser-agent explorer (Stagehand, Playwright) in AgentBay cloud sandboxes: a breadth scan, then parallel per-section depth agents; a coverage ledger with retry rounds; a risk gate before data-changing actions; hard payment blocks; and memory compaction for long runs.
- Built human login handoff over CDP (the user signs in on the same browser profile; the agent restores the session and continues). Explore/record/render run as Node.js/BullMQ jobs behind a multi-provider LLM gateway. 209 commits, ~500 tests.

Delphinus Lab

Nov 2024 – Jul 2026

Developer Relations Engineer → Ecosystem Director (Jun 2025) | Remote, concurrent with MPhil

- Rebuilt the zkWasm browser SDK from scratch (unified wallet context, L1/L2 signing and deposits), wrote the dApp-starter CLI, and maintained the TypeScript mini-rollup services.
- Solo-built staking, launchpad, prediction-market and social-binder apps, plus the zkWasm Hub site and developer docs; ran DevOps for 8+ ecosystem apps, which reached **thousands of users** and **multi-million-USD TVL**.
- Led the team that built a hosted Kubernetes deployment platform (GitHub integration, Helm jobs, async status/log tracking, CI/CD), hosting **10+ projects**.

Earlier roles (2021–2024, alongside undergraduate studies)

- Rosen** — Founding Full-Stack Engineer (2022–23): React / React Native apps for iOS, Android and Web; backed by PayPal and Circle. **2PM.Network** — Co-Founder & CEO (2024). **HashPunk** — Product Manager (2021–22), raised US\$1M. **1CIRCLE** — Full-Stack Engineer (2021–22). **Conflux** — Intern (2021).

Education

The Hong Kong University of Science and Technology (Guangzhou)

Sep 2024 – Oct 2026

Red Bird MPhil in Data Science and Analytics

The University of Edinburgh

Sep 2020 – Jun 2024

MA in Cognitive Science, First Class Honours | Exchange: Fudan University, Computer Science (2022–23)

Selected Projects

Viciking Production Tools

Aug – Oct 2026

In production | C# / .NET, SQLite; FastAPI, React, RF-DETR, TensorRT, FFmpeg

- **Unattended ad-platform uploader:** batch-uploads drama assets and submits them for review. Every irreversible write persists its intent and takes a one-time authorization first; lost receipts trigger read-only reconciliation, never a blind resubmit; per-file checkpoints skip published files on restart; failing dramas are quarantined while the queue continues.
- **GPU video redaction:** detects and masks 13 object classes in production videos. Built the Label Studio annotation loop, RF-DETR fine-tuning with TensorRT export, a lease-based job API, and a timeline review UI.

Novel-to-Script-Team — Multi-Agent Adaptation Harness

175 ★ 33 forks

Open source, sole author; stars grew through my own promotion | Agent Skills, Python, ChromaDB | for Claude Code / Codex

- Designed 17 role agents and 25 skills that take a novel through ingest, analysis, episode planning, writing, review and storyboard / video prompts. A review director gates every episode, and failures loop back for rewrite and re-review.
- Kept context bounded: the next phase is derived from which files exist, and per-project agent state lets work resume within an episode and reset between episodes to avoid context overflow. Hybrid ChromaDB + jieba TF-IDF retrieval grounds writing in reference scripts.

ShotLoom — AI Film Shot-Breakdown Workspace

24 ★

Open source, sole author, self-promoted | FastAPI, Redis/RQ, React, TypeScript, Qwen vision + ASR

- Built a shot-by-shot film breakdown: scene detection and clip cutting, then Qwen vision analysis of shot size, camera movement, lighting and narrative function, aligned with ASR transcripts and a whole-film continuity report.
- Routed long videos to whole-video or overlapping chunked passes with bounded model concurrency and retries; RQ workers with refresh-safe SSE progress; Excel/PDF export; 107 end-to-end tests.

RingChat — Agent-Native Messaging

Jan – Mar 2026

Team lead, client engineer | Kotlin / Jetpack Compose, Web, iOS | OpenClaw (self-hosted agent gateway)

- Led a **6-person team** building a cross-platform IM in which every user gets a personal AI agent, and worked mainly on the clients (now continued by another team as Gold House). On Android, built the chat UI, optimistic media upload, contact and group search, unread badges and multi-device sync.

Skills

Agents	Context engineering (cache-stable prefixes, compaction, file-backed state), tool/MCP design, multi-agent orchestration, browser agents (Stagehand, Playwright, CDP, AgentBay), human-in-the-loop, evals and LLM-as-judge
Frontend & mobile	TypeScript, React 19, TanStack (Start / Query), Next.js, Vite, Remotion; React Native, Kotlin / Jetpack Compose
Backend	Python (FastAPI), Node.js, Go; PostgreSQL / pgvector (SKIP LOCKED queues), Redis, Temporal, BullMQ, SSE, WebSocket
Infra & models	Docker, Kubernetes, CI/CD, OpenTelemetry; OpenAI, Anthropic, Gemini, Qwen, Doubao/Seedance via Vercel AI SDK and gateways
Other	Solidity, Rust, C++ (ESP32), PyTorch, scikit-learn

Publication

Xiaoxiao YU, L. Du, C. Zeng, L. Deng, H. Luo, X. Gao, J. Tang. *SocialAttributionQA: A Benchmark for Attribution-Evidence-Aware Social Web Question Answering*. **WISE 2026** (CCF B), accepted; Springer LNCS. First author: built a 1,200-question benchmark over 725k Mastodon posts and the ATLAS graph-plus-retrieval solver (**95.3%** accuracy), with ablations and error analysis.

Selected Hackathons & Honours

33 hackathon and competition awards (12 first place or champion); AI and full-stack highlights:

Ava Box — Voice-First Trading Agent

AVE CLAW Hackathon 2026 — Champion

Sole developer | ESP32-S3, C++, Python, LVGL

- Built a handheld voice agent that routes speech to trading tools based on the current screen; every order needs a button confirmation. Extracted into the open DeviceKit framework (pluggable chain/ASR/LLM/TTS adapters, OTA).

OutcomeX — AI Delivery & Settlement

HashKey Chain Horizon 2026 — 2nd, DeFi Track

Sole developer, 173 commits | FastAPI, AgentSkillOS (skill-DAG agent runtime), Solidity

- Built a marketplace where paid AI jobs run in an AgentSkillOS subprocess that streams DAG status, logs and artifacts back to the app, with escrow, refunds and revenue claims in 7 separate Foundry contracts.

2025	Zama Bounty, Universal FHEVM SDK — Winner	2024	BeWater AI Crypto — 1st, Security & Privacy
2025	ICBC Cup FinTech, Guangdong — First Prize	2024	NEAR “[REDACTED]” — AI Bounty (solo)
2024	ETH Shanghai — Overall Champion	2021	PaddlePaddle Anti-Fraud ML — 1st (solo)

Founded LambdaSoc (Edinburgh) and Lambda Web3 (**1,000+ members**); **7,000+** Xiaohongshu followers for engineering content. Languages: Chinese (native), English (professional).